Milestone-1

Project Context and motivation

1. Title: Optimizing UAV Deployment for Multi-UAV Cooperative Sensing.
2. Domain: Networking
3. Real-world context: UAVs (unmanned ariel vehicles) are used to detect signals from multiple targets over large area
4. Uncertainty: Noise power is not known. Signal strength varies with the direction of antenna and placement of UAVs.

Objective

Maximizing the sum detection probability of targets using multiple UAVs.

The transmitted signals from multiple targets are detected using Particle Swarm Optimization (PSO) based approach.

The optimization is done over the spatial deployment of UAVs and orientations of directional antennas.

End-to-end system pipeline

- Signals received from targets to UAVs.
- Eigenvalue-based detections is performed in UAVs.
- Local binary decisions generated.
- Sent to fusion center.
- Using OR logic final decision is made.

Major components: UAVs, eigenvalue-based detector and fusion center.

Data Flow: signal transmitted → UAV received → covariance matrix → eigenvalue based detection test → fusion center → final decision

Sources of Uncertainty in the Project

Noise measurement: noise is measured at UAV receivers using Additive White Gaussian Noise (AWGN).

Environmental Variability: distances between targets and UAVs and changing antenna alignment.

Modelling Uncertainty: fluctuations in noise power and variability in signal-to-noise ratio (SNR).

Key Random Variables and Their Roles :

Received signal samples

random signal emitted by kth target

Noise

noise at the u-th UAV receiver

test statistics

ratio of max to min eigen values

Eigen values

primary statistics used for decision making

Binary decision variables

D=1 for detected and D=0 for not detected

Probabilistic Models and Assumptions

Models used:

- Noise as zero-mean AWGN
- Test statistic approximated as gaussian for large samples

Assumptions:

- Fixed UAVs height
- Large sample size
- Noise is assumed independent and uniformly distributed.

Justification:

Gaussian approximation simplifies detection probability analysis.

Probabilistic Reasoning and Inference Logic

The probabilistic detection model defined earlier directly determines the objective function optimized in simulations.

Individual Detection

Each UAV computes its own detection probability

Cooperative Fusion

OR-rule combines independent decisions

Enhanced Reliability

More UAVs increase joint detection probability

Joint Detection Probability Formula :

$$P_{coop} = 1 - \prod (1 - P_d)$$

This expression demonstrates how cooperation reduces uncertainty. As more UAVs participate, the product term decreases, increasing overall detection probability.

Higher redundancy through multiple UAVs leads to improved system reliability.

Operationalising Probabilistic Reasoning

Simulation-Based Evaluation

Implementation through computational simulations comparing random versus optimised UAV deployment strategies.

Simplified Models

Feasibility considerations necessitate simplified channel and noise models while preserving essential probabilistic characteristics.

Key Metrics

- Cooperative detection probability (primary metric)
- False alarm rate control
- Convergence behaviour of optimisation algorithms
- Computational complexity analysis

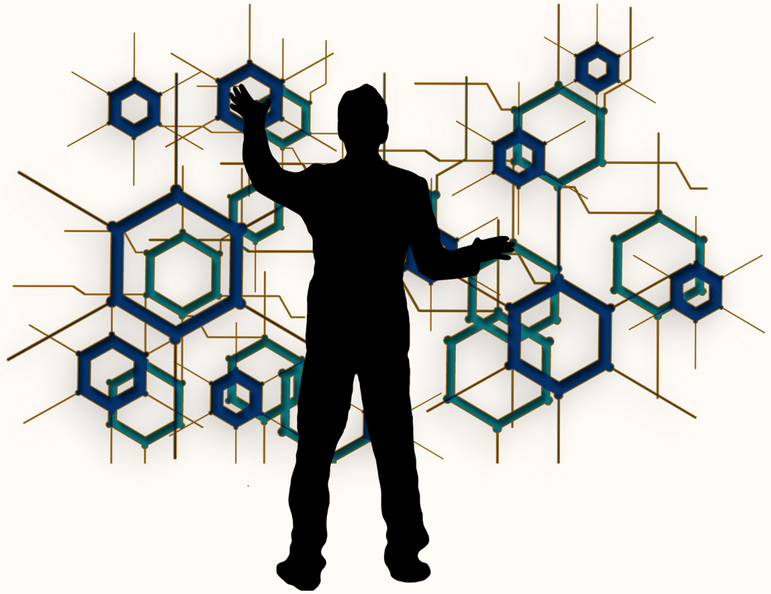
Current Limitations and Conceptual Gaps

The use of algorithms like PSO can become computationally expensive due to increased number of UAVs and targets.

The model assumes ideal communication between the UAVs and the fusion center.

This deployment optimization does not fully account for highly mobile targets.

Future Refinements and Role Coordination:



The project manager ensured the regular commits of all the members and took regular updates to ensure the task completion and coordination between two people working on one task.

Scale Up

Extend simulations to larger UAV networks

Refine Models

Improve noise and channel modelling

Validate Assumptions

Test independence and model accuracy

Code Management

GitHub integration for version control

Group level understanding

Using maximum-minimum eigenvalue test statistics, the system can achieve high detection performance even is low SNR or noise fluctuations.

Spatial deployment is the primary tool to enhance the statistical reliability of the binary decision variable.